Lifecycle Theory and Financial Restructuring of Distressed Firms

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Abstract

Firms can be divided into four lifecycle stages namely, Birth, Growth, Mature, and Decline. Firms in a lifecycle stage are characterized by similar size, ownership pattern, managerial control, capital structure, cash flows and future growth opportunities. It seems probable that such factors could affect how firms restructure in response to financial distress. This paper examines three kinds of financial restructuring strategies namely, Dividend cut, Net debt issuance and Net equity issuance and their relation with lifecycle stages for 2813 unique Indian firms for the period 2002 to 2015. I find that (a) financially distressed firms are less likely to cut dividends, but Mature and Decline distressed firms are more likely to cut dividends, (b) financially distressed firms are more likely to raise net debt but Birth and Growth distressed firms are less likely to raise net debt, and (c) financially distressed firms are less likely to raise net equity but Growth and Mature distressed firms are more likely to raise net equity than other lifecycle stage firms.

Keywords: Lifecycle of firms, Financial Distress, Financial Restructuring

JEL Classification: G30, G32, G39

1 Introduction

Lifecycle theory suggests that firms, like products or marketing strategies in other disciplines, also undergo four different lifecycle stages. These four stages are defined as Birth, Growth, Mature and Decline. Firms in a lifecycle stage have similar size, ownership, managerial control, capital structure, cash flows and future growth opportunities (Miller and Friesen, 1984). Such differences could affect how a firm reacts to financial distress. Typically, a firm undergoing financial distress would undertake different types restructuring or preventive measures, including financial restructuring (Sudarsanam and Lai, 2001). I consider three such financial restructuring activities, namely, cutting dividends, raising net debt, or raising net equity to explore the

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effects of lifecycle stages on restructuring choices by distressed firms. Particularly, I examine if firms in different lifecycle stages differ significantly in strategies of financial restructuring. My study has consequences for managers. It is important that the firm managers take right types of preventative measures in light of financial distress. These preventive strategies should be appropriate for the stage of the lifecycle that the firm is in (Koh et al., 2015). As per my knowledge, this is the first study in this context for Indian companies. Therefore, the novelty and importance of restructuring in financially distressed Indian firms form the motivations of this paper.

The paper proceeds as follows. Section 2 reviews relevant literature related to lifecycle theory and restructuring, and develops the hypotheses. Section 3 defines the data and the variables. Results and analysis are shown in section 4. Section 5 performs some additional tests. Section 6 concludes.

2 Literature Review

Prior literature suggests that as a firm grows old and mature, the management and shareholder focus shifts. Based on differences in a firm's size, ownership, managerial control, capital structure, cash flows and future growth opportunities, it can be classified into one of four lifecycle stages - Birth, Growth, Mature, and Decline. A typical firm's lifecycle is described in Mueller (1972). An entrepreneur with a good, new and viable idea, skill or insight decides to start a firm. The innovating entrepreneur uses her own capital and external financing (in exchange for a promise to repay the debt with interest or to share profits in case of equity) to bring the idea into a working form. This initial capital comes at higher costs as the uncertainity of future cash flows is higher for a new firm. This stage is akin to the 'Birth' of the firm. If the idea and the firm are successful, the firm expands. The stockholders reinvest most of their profits into the firm and, even increase their capital invested to gain from the new and better growth opportunities. This stage is referred to as 'Growth' of the firm. As the firm grows old and competition develops, these growth opportunities begin to fall. At this point, the stockholders prefer to reinvest only a part of profits, and distribute the rest as dividends to maximize their welfare. As the market develops further - through newer technologies or increased competition - the firm begins to put a larger part of its earnings into dividends. Mueller (1972) names this stage as 'Mature'. Eventually, when all of the capital stock of the firm is used up, the firm pays almost the entire cash flow as dividends. At this point, the firm is said to be in 'Decline'.

Firms in different lifecycle stages have different characteristics. Firms in Birth lifecycle stage tend to be smaller, have more concentrated ownership structure consisting primarily of the entrepreneurs and focus highly on research and innovations (Miller and Friesen, 1984; Anthony and Ramesh, 1992). For such firms, the uncertainty regarding future growth is generally higher, and they are characterized by high market-to-book ratios (Pástor and Pietro, 2003). Firms in the Growth stage are medium -sized and start defining separate control and ownership rights (Miller and Friesen, 1984). These studies argue that firms in later stages of Maturity and Decline are less focused on innovation than Birth and Growth firms. Mature firms look for smooth earnings generating operation in well-defined markets, while Decline firms are characterized by stagnation or declining profits (Miller and Friesen, 1984). These firms aim to extract as much wealth from current operations as possible. Hence, it is observed that firms in various stages of lifecycle have contrasting growth and capital capacities (Anthony and Ramesh, 1992).

Given such differences between firms characteristics, focus and future outlooks in the lifecycle stages, it seems probable that these firms also differ in how they react and restructure in response to situations of financial distress. Typically, the literature points to four kinds of restructuring a financially distressed firm can undertake. These are managerial, operational, asset and financial (Sudarsanam and Lai, 2001; Koh et al., 2015). Managerial restructuring refers to replacing the top managerial positions like CEOs when the firm performance declines. Operational restructuring includes reducing costs of operations and other forms of raising cash through selling property or plant. However, Sudarsanam and Lai (2001) stress that such operation strategies help improve distress only in the short-term. Asset restructuring is defined as selling off non-core lines of business to reduce unrelated diversification and concentrate on the core strengths of the firms. Recent evidence observes that asset restructuring can be value-adding (Atanassov and Kim, 2009). Finally, financial restructuring includes changes in dividend policy or capital structure of the firm, including the net issuance of debt, and net issuance of equity. While Sudarsanam and Lai (2001) include dividend as part of equity restructuring, Koh et al. (2015) count it separately. I follow the latter approach.

This paper aims to study the different restructuring activities that financially distressed firms in different lifecycle undertake. Such studies have been recently done for the US markets (Koh et al., 2015). However it seems that the current trend has not been studied in India. Since the scope of my study is limited, I study how firms' lifecycle stage constraints a firm's policy of financial restructuring. In particular, I explore how dividend-cut related financial restructuring performs with respect to net debt and net equity issuance financial restructuring for different lifecycle stage firms in financial distress. This aims to establish whether some kinds of firms are more likely to do one kind of financial restructuring than the rest.

2.1 Hypotheses Development

Dividend Cut: There are two contrasting evidence on dividends. On one hand, one would expect that financial distress will be positively related to dividend cuts as firms undergoing distress would want to cut on cash flow divestment and reinvest all the profits into the firm itself (according to the Pecking-order theory by Myers and Majluf (1984) that says firms prefer internal financing like dividend cuts followed by debt and then by equity to raise capital). On the other hand, evidence on persistence of dividends is also highly documented. Graham et al. (2015) show that reducing dividends signals persistent bad future-times to the market and subsequent reputation reduction. In other words, they find evidence in support of information content of dividends and hence one would expect that financial distress will be negatively related to dividend cuts. In light of such contrasting evidence, I test which of the above contrasting strategies for dividend are more common in Indian firms (H1a) during distress.

Further, I expect that Mature and Decline firms are more likely to cut dividends (H1b). The seminal works of DeAngelo and DeAngelo (1990) find that larger firms in NYSE tend to cut dividends during distress during the period 1980-1985. The reasons for dividend cut include strategic considerations and aversion to completely omit dividends to maintain long history of continuous dividends. This is also in line with results found by Koh et al. (2015). Additionally, many firms in early stages like Birth and Growth are less likely to pay dividends in the first place and prefer to reinvest funds in the firm (Miller and Friesen, 1984).

Net Debt Issue: I expect that debt issuance would be positively correlated with financial distress (H2a). There are two reasons to expect this: First, financial distress generally results in inadequate supply of equity, forcing firms to borrow debt which ensures easier monitoring and allows opportunities for renegotiation than dispersed sources of equity (**?**). Second, according to the Pecking-order theory of capital structure (Myers and Majluf, 1984) which shows that a rational firm would issue debt before it issues equity during need of external capital.

However, I expect that Birth and Growth distressed firms would be less likely to issue debt (H2b), in line with **?** work. Empirical evidence by **?** suggests that smaller growing firms sway away from Pecking-order theory. Debt issuance for early stage firms would usually come with strict covenants and higher monitoring clauses that such firms find restricting in future. In same line of argument, Decline firms are most likely to issue debt in financially distressed times (H2c). Dwindling future growth opportunities may make it harder for such firms to get any equity, while it may be easier for them to get debt using any prior reputations. Hence, Decline firms may find lenders relatively easily and at a lower cost causing them to prefer debt over equity as Pecking-order predicts.

Net Equity Issue: I also expect that issuance of equity would be negatively related to financial distress according to the Pecking-order theory (H3a). Equity is considered restructuring of the last resort (Myers and Majluf, 1984).

Additionally, I expect that Growth firms will be more likely to issue equity (H3b) (a consequence of H2b). This is in accordance to the seminal work by **?** who show that small growing firms prefer equity over debt as they may save on tax shields and be protected from strict covenants in their early stages.

3 Data and Variables

Data for all firms listed on BSE and NSE (5560 firms) was taken from Centre for Monitoring Indian Economy (CMIE) Prowess dx database. The information included details from annual standalone financial statements, market capitalization, ownership and history of firms. The time period for the study is 2001 to 2015. This ensures that financial distress time periods of 2001-02 Crisis and 2008-09 Global Crisis are included. Financial, Insurance, and Public Utilities Firms have been excluded as in the literature (capital structure is governed by different rules for such firms). After variables computation and dropping observations with missing data, I am left with 22576 unique firm-year observations, which includes 2813 unique firms and 14 unique years (2002 to 2015). Since some firms were listed during the sample period, I have an unbalanced panel dataset. The total firms in each year are shown in Table 1. As expected, the number of firms have increased with time.

3.1 Lifecycle Stages

We use Anthony and Ramesh (1992)'s (also folllowed by Koh et al. (2015)) methodology to classify firms into Birth, Growth, Mature or Decline lifecycle stages. I use four measures for each firm:

1. Annual Dividend as a percentage of Income before Extraordinary Items (DP)

DP_t = Equity Dividend in year t/Income before extraordinary items and discontinued operations in year t

2. Sales Growth in percent (SG)

   SG_t = Net sales in year t/Net sales in year t-1

3. Capital Expenditure as a percentage of Total Value of the firm (CEV)

   CEV_t = Capital expenditure in year t / (Market Capitalization + Book Value of Assets) in year t

4. Age of the firm (Age)

   Age_t = t - Year of incorporation

For each year, I first calculate all four measures for each firm. Then year-wise, I separate firms into quartiles for each of the four measures calculated above. The next step is to assign scores. Scores of (1,2,3,4), (4,3,2,1), (4,3,2,1) and (1,2,3,4) are assigned from lowest to highest quartiles for Dividend Payout, Sales Growth, Capital Expenditure Value and Age respectively. This is done so that, for example, Birth firms characterized by lowest dividend payout, highest sales growth, highest capital expenditure and lowest age get a total ex-ante lowest score of 4 (=1*4). Similarly, a score of 2*4 for Growth firms, 3*4 for Mature and 4*4 (highest score) for Decline firms. To get the final stage of the lifecycle of each firm for each year, I sum the four scores from the four measures, and again divide this sum into quartiles. The firms in the first (lowest), second, third and last (highest) quartile are separated as Birth, Growth, Mature and Decline firms, respectively.

For each year, the number of Indian firms in each lifecycle stage is shown in Table 1. The number of firms has increased with time, with a minor drop in the recent period. This could be due to dropping off some missing observations.

3.2 Financial Distress

Definition of Financial Distress is taken from Asquith et al. (1991). Financial Distress (FD) for a firm i in year t is defined as an indicator variable that takes value 1 if the firm's PBITDA (Profit Before Interest, Tax, Depreciation, Amortization) over reported interest expenses is less than 1 in year t and year t-1 (2 years consecutively), OR is less than 0.8 in single year t (applied on firms which were non-distressed after '2 years consecutively' definition). Firms with ratio between 0.8 and 1 in one year are not counted as financially distressed. Methods used by Sudarsanam and Lai (2001) and Koh et al. (2015) based on Taffler's Z-score and Distance-to-default (Merton, 1974) can also be used. Note that I used data for year 2001 and 2002 to see if a firm is distressed in year 2002.

The number of distress firms in each year is reported in Table 1. The number of distressed firms has grown in the recent times, but this could be accounted by the fact that the total number of

Table 1: Number of Firms in each Lifecycle Stage

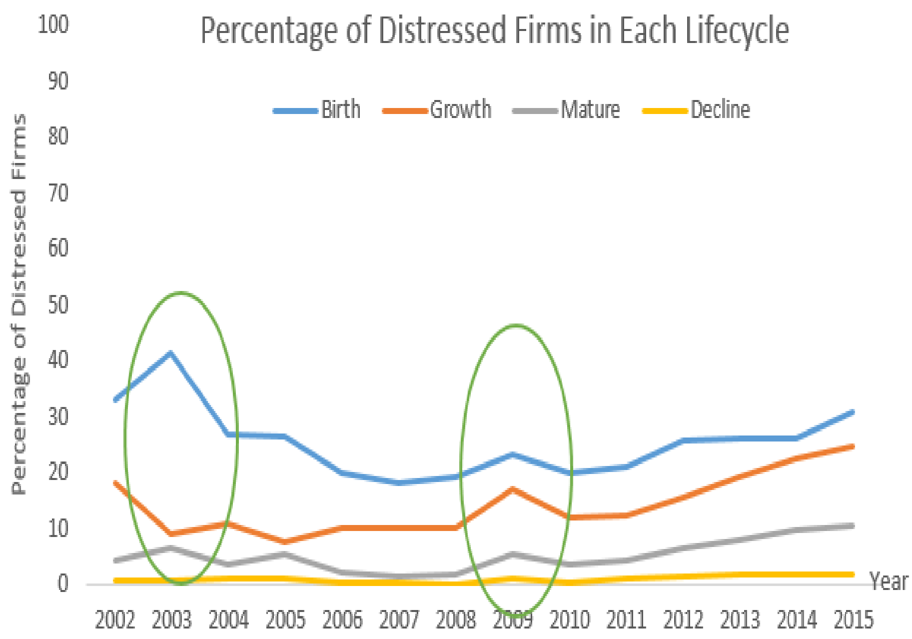
The lifecycle stage is calculated based on four measures, namely Annual dividend as a percentage of Income, Sales Growth in percent, Capital Expenditure as a percentage of Total Value of the firm, Age of the firm. Distress in year t is defined 1 if a firm's PBITDA over reported interest expenses is less than 1 in year t and year t-1, OR is less than 0.8 in year t.

Year	Total Firms	Distressed	Birth	Growth	Mature	Decline
2002	572	80	142	143	144	143
2003	705	101	177	176	176	176
2004	939	98	234	235	235	235
2005	1244	124	309	312	312	311
2006	1506	120	375	377	377	377
2007	1688	125	420	423	422	423
2008	1796	137	449	449	449	449
2009	1776	204	444	444	444	444
2010	2063	182	515	515	517	516
2011	2065	196	517	515	517	516
2012	2086	253	520	522	522	522
2013	2067	282	517	517	517	516
2014	2031	303	508	508	508	507
2015	2038	343	510	509	509	510
Total	22576	2548	5637	5645	5649	5645

firms has also increased over the years. Distressed firms as a percentage of total firms in each lifecycle stage in each year are plotted in Figure 1. Birth lifecycle stage firms have the highest percentage of distressed firms, which seems intuitive as these firms on account of uncertain future are generally financially constrained. Similarly, Decline firms have the least percentage of distressed firms as these firms generally have few opportunities and hence expenses. The graph shows (green circles) the years of Financial Crisis of 2001-02 and 2008-09 where the number of distressed firms has risen, as is expected.

Figure 1: Percentage of Distressed Firms in each Lifecycle Stage

The figure shows percentage of distressed firms in each lifecycle stage in each year. It is computed as Number of Distressed Firms in Lifecycle Stage k in year t / Number of Firms in Lifecycle Stage k in year t, from 2002 to 2015 (k is either Birth, Growth, Mature or Decline). The lifecycle stage is calculated based on four measures, namely Annual dividend as a percentage of Income, Sales Growth in percent, Capital Expenditure as a percentage of Total Value of the firm, and Age of the firm. Distress in year t is defined 1 if a firm's PBITDA over reported interest expenses is less than 1 in year t and year t-1, OR is less than 0.8 in year t



3.3 Financial Restructuring

Following Chen and Zhang (1998) and Koh et al. (2015), I define financial restructuring variables for firm i at time t as indicator variables that take value 1 if:

1. Dividend Restructuring (Div_restr): Total dividends in year t decrease by more than 25% with respect to year t-1
2. Debt Restructuring (Debt_restr): The firm's net debt issued exceeds 5% of the book value of its total assets at year t, where net debt = issuance of long-term debt in year t less repayment of long debt in year t
3. Equity Restructuring (Equity_restr): The firm's net equity issued exceeds 5% of the book value of its total assets at year t, where net equity = issuance of shares in year t less repurchase of shares in year t

3.4 Control Variables

I use the following firm-level control variables for year t:

1. Tobin's Q = Market Value in year t/Book Value of assets in year t
2. ln(Totalassets)= log of Total Assets in year t
3. PctInstitution = Percentage Institutional Investor holdings in year t
4. Leverage = (Current Liability + Long Term debt) in year t /(Market Capitalization + Book Value Assets) in year t
5. CFO = Cash Flow from operations in year t/ Total Assets in year t

Additionally, I also control for Industry Fixed-Effects through first two digits of NIC codes, and Year Fixed-Effects. I have 20 industry dummies and 14 year dummies.

4 Results

I run a Logit, Random-effects Regression model on panel data following Koh et al. (2015). I run three separate models where the dependent variable is taken as Dividend restructuring, Debt restructuring, and Equity restructuring. The base cases for the regressions are reported keeping in mind ease of interpretation, although all specifications give qualitatively similar results.

4.1 Dividend Restructuring

The regression equation with base as Birth Distressed firms is as follows:

$$\begin{aligned} Dividend\_Restructuring_{it} = {} & \alpha_1 + \alpha_2 Growth_{it} + \alpha_3 Mature_{it} + \alpha_4 Decline_{it} + \alpha_5 FD \\ & + \alpha_6 Growth * FD_{it} + \alpha_7 Mature * FD_{it} + \alpha_8 Decline * FD_{it} \\ & + \alpha_9 TobinsQ_{it} + \alpha_{10} ln(Totalassets)_{it} + \alpha_{11} PctInstitution \\ & + \alpha_{12} Leverage + \alpha_{13} CFO_{it} + IndustryFE + YearFE \end{aligned}$$

The variables are as defined in Section 3 above. Birth, Growth, Mature and Decline are indicator variables that are 1 if a firm is in Birth lifecycle, Growth lifecycle, Mature lifecycle or Decline lifecycle respectively. I have also included interactions between lifecycle variables and financial distress indicator variable. Coefficients of interest are $\alpha_5, \alpha_6, \alpha_7, \alpha_8$.

The results for Dividend Restructuring are shown in Table 2. The coefficient of Financial Fistress (FD) is negative and significant at 1% implying that a firm in financial distress is less likely to cut dividends. This supports thoery of persistence of dividends (Graham et al., 2015) and our hypothesis H1a, but contradicts the conclusions by Mueller (1972) and Myers and Majluf (1984). It seems that information content of the dividend is stronger than maintaining internal capital during distress.

However, for Mature X FD, that is Mature firms in distress, the coefficient is positive and significant and log odds ratio is exp(1.189153) = 3.284, implying that distressed firms in Mature lifecycle stage are over thrice as likely than Birth firms in distress to cut dividends, all other factors being equal. Similarly, the log odds ratio for Decline firms in distress is exp(1.275399) = 3.58 and significant implying that distressed firms in the Decline lifecycle stage are almost

Table 2: Dividend Restructuring

The dependent variable is Dividend Restructuring for firm i, time t which is defined as an indicator variable that takes value 1 if the total dividends by firm i in year t decrease by more than 25% with respect to year t-1. *** denotes significance at 1%, ** denotes significance at 5%, * denotes significance at 10%.

Dividend_Restructuring	Coefficient	Standard Error	P-value
Constant	-3.190223	0.216443	0.000***
Growth	0.0106284	0.0680714	0.876
Mature	-0.089162	0.070027	0.203
Decline	-0.2815639	0.0750843	0.000***
FD	-0.9046699	0.1333811	0.000***
Growth*FD	0.2237264	0.1927797	0.246
Mature*FD	1.189153	0.2177413	0.000***
Decline*FD	1.275399	0.4427463	0.004***
Tobin's Q	-0.0073163	0.0116332	0.529
Log(Total Assets)	0.276164	0.0177594	0.000***
Institutional Holding	-0.0164645	0.0026434	0.000***
Leverage	-0.3398206	0.1074746	0.002***
Cash Flow from Operations	-0.1533514	0.0953595	0.108

3.5 times as likely than Birth firms in distress to cut dividends. Similarly, Growth X FD coefficient is also positive, although not significant. Thus, in line with evidence that larger firms are more likely to cut dividends (DeAngelo and DeAngelo, 1990) and supports our hypothesis H1b. It could be that Mature and Decline firms are less likely to take into account the negative signalling effect of reducing dividends, probably because their future opportunities are inherently limited (Miller and Friesen, 1984) and stockholders want to consume as much of cash flows as possible.

Also, other firms in Birth and Growth stage may not pay dividends in the first place (Miller and Friesen, 1984). I observe that 5105 out of 5637 (about 90%) observations in Birth stage do not pay any dividends. This figure for Growth, Mature and Decline lifecycle stages is about 61%, 33% and 8%, respectively.

4.2 Debt Restructuring

The regression equation with base as Decline distressed firms is as follows:

$$\begin{aligned} Debt\_Restructuring_{it} &= \alpha_1 + \alpha_2 Birth_{it} + \alpha_3 Growth_{it} + \alpha_4 Mature_{it} + \alpha_5 FD \\ &+ \alpha_6 Birth * FD_{it} + \alpha_7 Growth * FD_{it} + \alpha_8 Mature * FD_{it} \\ &+ \alpha_9 TobinsQ_{it} + \alpha_{10} ln(Totalassets)_{it} + \alpha_{11} PctInstitution \\ &+ \alpha_{12} Leverage + \alpha_{13} CFO_{it} + IndustryFE + YearFE \end{aligned}$$

The variables are as defined in Section 3 above. Birth, Growth, Mature and Decline are indicator variables that are 1 if a firm is in Birth lifecycle, Growth lifecycle, Mature lifecycle or Decline lifecycle respectively. I have also included interactions between lifecycle variables and financial distress indicator variable. Coefficients of interest are $\alpha_5, \alpha_6, \alpha_7, \alpha_8$.

Table 3: Debt Restructuring

The dependent variable is Debt Restructuring for firm i, time t which is defined as an indicator variable that takes value 1 if the firm i's net debt exceeds 5% of the book value of its total assets at year t, where net debt = issuance of long-term debt in year t less repayment of long debt in year t. *** denotes significance at 1%, ** denotes significance at 5%, * denotes significance at 10%.

Debt_Restructuring	Coefficient	Standard Error	P-value
Constant	-2.912915	0.2956334	0.000***
Birth	-0.0065662	0.0814118	0.936
Growth	0.069343	0.0684718	0.311
Mature	0.1122789	0.0606531	0.064*
FD	0.9766371	0.3713206	0.009***
Birth*FD	-1.180369	0.38448	0.002***
Growth*FD	-1.103035	0.3864786	0.004***
Mature*FD	-1.263035	0.4080161	0.002***
Tobin's Q	-0.0393094	0.0201989	0.052*
Log(Total Assets)	0.2829877	0.0230628	0.000***
Institutional Holding	-0.0046339	0.002999	0.122
Leverage	0.0455619	0.1252348	0.716
Cash Flow from Operations	-0.1217512	0.0973438	0.211

The results for Debt Restructuring are shown in Table 3. The variable for financial distress (FD) is positive and significant at 1% implying that a firm in financial distress is more likely to undergo debt restructuring. This is in support to results in US reported by Koh et al. (2015) and Pecking-Order Theory (Myers and Majluf, 1984). This allows us to not reject our hypothesis H2a. Hence, Indian companies in distress prefer debt over equity. Financial distress could mean inadequate supply of equity, forcing firms to borrow debt which ensures easier monitoring by the lenders than dispersed sources of equity (**?**).

However, log odds ratios of Birth X FD and Growth X FD, that is Birth firms in distress and Growth firms in distress are exp(-1.180369)= -0.307 and exp(-1.103035) = -0.33 respectively. Both are significant at 1% implying that distressed firms in Birth and Growth lifecycle stage are only approximately 31% as likely as Decline firms in distress to raise debt, all other factors being equal. I am not able to reject H2b. It seems that firms in early stages find debt's stricter covenants and higher monitoring restricting. In same line of argument, we see that Mature X FD are also significantly less likely than Decline firms in distress to issue debt. All these together can imply that Decline distressed firms are most likely to issue debt in financially distressed times, supporting H2c. It seems that managers of Decline distressed firms may get easier debt using any prior reputation benefits. Hence, Decline firms may find lenders at relatively lower costs, even during financial distressed times. Also, strict covenants would matter less to Decline distress firms as there are fewer projects in future (Miller and Friesen, 1984).

4.3 Equity Restructuring

The regression equation with base as Birth Distressed firms is as follows:

$$\begin{aligned}Equity\_Restructuring_{it} = \alpha_1 &+ \alpha_2 Growth_{it} + \alpha_3 Mature_{it} + \alpha_4 Decline_{it} + \alpha_4 FD \\ &+ \alpha_6 Growth * FD_{it} + \alpha_7 Mature * FD_{it} + \alpha_8 Decline * FD_{it} \\ &+ \alpha_9 TobinsQ_{it} + \alpha_{10} ln(Totalassets)_{it} + \alpha_{11} PctInstitution \\ &+ \alpha_{12} Leverage + \alpha_{13} CFO_{it} + IndustryFE + YearFE\end{aligned}$$

The results for Equity Restructuring are shown in Table 4. The variable for financial distress (FD) is negative and significant at 1% implying that a firm in financial distress is less likely to undergo equity restructuring and raise equity. The result is in support of hypothesis H3a and Pecking-order theory (Myers and Majluf, 1984). Equity issuance is not a preferred restructuring strategy.

For Growth X FD, that is Growth firms in distress, the coefficient is positive and significant at 5%, and the log odds ratio is exp(0.5109143) = 1.66, implying that distressed firms in the Growth lifecycle stage are 160% likely than Birth firms in distress to issue equity, all other factors being equal. Thus, H3b is not rejected. This is in accordance to the seminal work by **?** who show that smaller growing firms prefer equity over debt. These may save on tax shields and be protected from strict covenants of debt (which is common for firms in their early stages) that may prove unviable in future.

However, the log odds ratio for Mature firms in distress is also positive and significant at 1%. It is exp(1.133547)= 3.1, implying that distressed firms in the Mature lifecycle stage are almost 300% likely than Birth firms in distress to raise equity, all other factors being equal. I can not think of a reason for this observation. Further research examining such observation is being explored.

5 Additional Tests

I perform three robustness checks. First, I check if Financial Restructuring itself is significant during distress. I create a dummy variable 'Financial Restructuring' for each firm i in year t. It is assigned a value of 1 if the firm i in year t engages in either kind of financial restructuring, that is Financial Restructuring is 1 if the firm engages in either Dividend Restructuring, or Debt Restructuring or Equity Restructuring. It is 0 only when there is none of Dividend Restructuring, Debt Restructuring and Equity Restructuring by the firm i in year t.

The regression equation with base as Decline Distressed firms is as follows:

$$\begin{aligned}Financial\_Restructuring_{it} = \alpha_1 &+ \alpha_2 Birth_{it} + \alpha_3 Growth_{it} + \alpha_4 Mature_{it} + \alpha_4 FD \\ &+ \alpha_6 Birth * FD_{it} + \alpha_7 Growth * FD_{it} + \alpha_8 Mature * FD_{it} \\ &+ \alpha_9 TobinsQ_{it} + \alpha_{10} ln(Totalassets)_{it} + \alpha_{11} PctInstitution \\ &+ \alpha_{12} Leverage + \alpha_{13} CFO_{it} + IndustryFE + YearFE\end{aligned}$$

The results for Financial Restructuring are shown in Table 5. The coefficient of financial distress (FD) is positive and significant at 5% implying that a firm in financial distress is likely to undergo some form of financial restructuring. The coefficients of Birth X FD, Growth X FD

Table 4: Equity Restructuring

The dependent variable is Equity Restructuring for firm i, time t which is defined as an indicator variable that takes value 1 if the firm i's net equity exceeds 5% of the book value of its total assets at year t, where net equity = issuance of shares in year t less repurchase of shares in year t. *** denotes significance at 1%, ** denotes significance at 5%, * denotes significance at 10%.

Equity_Restructuring	Coefficient	Standard Error	P-value
Constant	-3.212816	0.3350729	0.000***
Growth	-0.1412297	0.0860066	0.101
Mature	-0.2888686	0.0908971	0.001***
Decline	-0.5646288	0.1031758	0.000***
FD	-0.3973898	0.1390694	0.004***
Growth*FD	0.5109143	0.2094162	0.015**
Mature*FD	1.133547	0.2724641	0.000***
Decline*FD	0.5527515	0.7818374	0.48
Tobin's Q	-0.0244537	0.0052617	0.000***
Log(Total Assets)	0.0891228	0.0233483	0.000***
Institutional Holding	0.0148483	0.0032722	0.000***
Leverage	-2.232935	0.1591653	0.000***
Cash Flow from Operations	-1.219842	0.1601554	0.000***

and Decline X FD are all significant. Decline firms are more likely to do financial restructuring than other lifecycle stages.

Second, to check if results are not driven by few firms going into distress multiple times I run the regressions again dropping firms from the sample the first time they become distressed. If a firm doesn't experience distress in any year it always remains in the sample. In such case, the number of observations drops to 18779. I report results for financial distress Table 6 (Dividend, Debt and Equity not reported for brevity - can be requested from author).

The results remain qualitatively unchanged. The coefficient of financial distress (FD) is positive and significant at 5% implying that a firm in financial distress is likely to undergo some form of financial restructuring and that this result is not driven by few firms going into financial distress more than once.

Table 5: Financial Restructuring

The dependent variable is Financial Restructuring for firm i, time t which is defined as an indicator variable that takes value 1 if the firm i engages in either Dividend Restructuring, Debt Restructuring or Equity Restructuring year t. *** denotes significance at 1%, ** denotes significance at 5%, * denotes significance at 10%.

Financial_Restructuring	Coefficient	Standard Error	P-value
Constant	-2.095092	0.2098833	0.000***
Birth	0.3237113	0.0634781	0.000***
Growth	0.2721749	0.0550406	0.000***
Mature	0.19895	0.0496191	0.000***
FD	0.6945294	0.3445062	0.044**
Birth*FD	-1.22737	0.3538237	0.001***
Growth*FD	-0.9365345	0.3559192	0.009***
Mature*FD	-0.608909	0.3709054	0.10*
Tobin's Q	-0.0073595	0.0040896	0.072*
Log(Total Assets)	0.3135442	0.0165607	0.000***
Institutional Holding	-0.0072412	0.0023487	0.002***
Leverage	-0.5484634	0.0916094	0.000***
Cash Flow from Operations	-0.5402054	0.1016276	0.000***

Table 6: Financial Restructuring with Dropped Observations

The dependent variable is Financial Restructuring for firm i, time t which is defined as an indicator variable that takes value 1 if the firm i engages in either Dividend Restructuring, Debt Restructuring or Equity Restructuring year t. *** denotes significance at 1%, ** denotes significance at 5%, * denotes significance at 10%.

Financial_Restructuring	Coefficient	Standard Error	P-value
Constant	-2.174027	0.2231049	0.000***
Birth	0.3997953	0.0682744	0.000***
Growth	0.3150441	0.0592943	0.000***
Mature	0.2173651	0.0536838	0.000***
FD	1.029106	0.4943403	0.037**
Birth*FD	-1.584517	0.5034363	0.002***
Growth*FD	-1.170503	0.5091894	0.022**
Mature*FD	-1.031954	0.532713	0.053*
Tobin's Q	-0.0096012	0.004548	0.035**
Log(Total Assets)	0.3201936	0.0183672	0.000***
Institutional Holding	-0.008011	0.0025471	0.002***
Leverage	-0.4687057	0.102658	0.000***
Cash Flow from Operations	-0.6275273	0.1294323	0.000***

Third, I see effects of Earnings Management on lifecycle and financial restructuring. It could be that a firm may manage its earnings to report PBITDA marginally higher than its interest expenses, so as to not be counted a 'distressed'. I therefore update the definition of financial distress to account for such cases. Financial Distress (FD) for a firm i in year t is now defined as an indicator variable that takes value 1 if the firm's PBITDA over reported interest expenses is less than 1.1 in year t and year t-1 (2 years consecutively), OR is less than 0.8 in single year t. The number of observations (22576) and results remain unchanged. A more extensive examination of how Earnings Management could affect our result are being studied in a follow-up paper.

6 Conclusion

Firms can be divided into four lifecycle stages namely, Birth, Growth, Mature, and Decline. Each lifecycle stage is characterized by similar size, ownership pattern, managerial control, capital structure, cash flows and future growth opportunities. Firms in birth lifecycle stage tend to be smaller, have more concentrated ownership structure, have uncertainty regarding future growth. Firms in the growth stage are medium-sized and start to define the separation of control and ownership rights. Mature firms look for smooth earnings generating operation in well-defined markets while decline firms are characterized by stagnation or declining profits and aim to extract as much wealth from current operations as possible. Such factors affect how firms restructure in response to financial distress. This paper examines three kinds of financial restructuring strategies namely, dividend cut, debt issuance and equity issuance and their relation with lifecycle stages for Indian firms.

I find that financially distressed firms are less likely to cut dividends, supporting the persistence of dividends studies but contradicting the Pecking-order theory. However, Mature and Decline distressed firms are more likely to cut dividends. Financially distressed firms are more likely to raise net debt, which is in line with conclusions Myer and Majluf's Pecking Order theory. However, Birth and Growth distressed firms are less likely to raise net debt which could be attributed aversion to stricter debt covenants that may prove detrimental in future. Finally, financially distressed firms are less likely to raise net equity, again in accordance with the Pecking-order Theory. In these, Growth and Mature distressed firms are more likely to issue net equity due to non-preference towards stricter debts.

There are two possible extensions of the paper. One can examine managerial, operational and asset restructuring along with financial restructuring. Secondly, one could use recent methods of defining financial distress. These extensions are currently being pursued by the author.

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